

Why Poorer Players Shouldn't Play the Same Board As Rich Players

Parameters

g_t = gold at time t , s_t = spending at time t , y_t = gold income at time t .

Assume the Intertemporal Budget Constraint is

$$g_{t+1} = (1 + r)g_t + y_t - s_t,$$

where r is the return to saving gold.

Rearranging

$$g_{t+1} = (1 + r)g_t + y_t - s_t.$$

Solve for s_t

$$s_t = (1 + r)g_t + y_t - g_{t+1}.$$

Divide both sides by $(1 + r)^t$:

$$\frac{s_t}{(1 + r)^t} = \frac{(1 + r)g_t}{(1 + r)^t} + \frac{y_t}{(1 + r)^t} - \frac{g_{t+1}}{(1 + r)^t}.$$

Simplify:

$$\frac{s_t}{(1 + r)^t} = \frac{g_t}{(1 + r)^{t-1}} + \frac{y_t}{(1 + r)^t} - \frac{g_{t+1}}{(1 + r)^t}.$$

Sum from $t = 0$ to T :

$$\sum_{t=0}^T \frac{s_t}{(1 + r)^t} = \sum_{t=0}^T \frac{g_t}{(1 + r)^{t-1}} + \sum_{t=0}^T \frac{y_t}{(1 + r)^t} - \sum_{t=0}^T \frac{g_{t+1}}{(1 + r)^t}.$$

expanding it gives

$$\sum_{t=0}^T \frac{g_t}{(1 + r)^{t-1}} = (1 + r)g_0 + g_1 + \frac{g_2}{1 + r} + \cdots + \frac{g_T}{(1 + r)^{T-1}},$$

and

$$\sum_{t=0}^T \frac{g_{t+1}}{(1 + r)^t} = g_1 + \frac{g_2}{1 + r} + \cdots + \frac{g_{T+1}}{(1 + r)^T}.$$

Subtracting

$$\sum_{t=0}^T \frac{g_t}{(1+r)^{t-1}} - \sum_{t=0}^T \frac{g_{t+1}}{(1+r)^t} = (1+r)g_0 - \frac{g_{T+1}}{(1+r)^T}.$$

Result

$$\sum_{t=0}^T \frac{s_t}{(1+r)^t} = (1+r)g_0 + \sum_{t=0}^T \frac{y_t}{(1+r)^t} - \frac{g_{T+1}}{(1+r)^T}.$$

If we impose the terminal condition

$$\frac{g_{T+1}}{(1+r)^T} = 0,$$

Then it becomes

$$\sum_{t=0}^T \frac{s_t}{(1+r)^t} = (1+r)g_0 + \sum_{t=0}^T \frac{y_t}{(1+r)^t}.$$

Define lifetime gold by

$$W \equiv (1+r)g_0 + \sum_{t=0}^T \frac{y_t}{(1+r)^t}.$$

Then

$$\sum_{t=0}^T \frac{s_t}{(1+r)^t} = W.$$

So now the players need to optimize by spending for maximum utility:

$$\max_{\{s_t\}_{t=0}^T} \sum_{t=0}^T \beta^t u(s_t)$$

Subject to

$$\sum_{t=0}^T \frac{s_t}{(1+r)^t} = W.$$

Grab the Lagrangian::

$$\mathcal{L} = \sum_{t=0}^T \beta^t u(s_t) + \lambda \left(W - \sum_{t=0}^T \frac{s_t}{(1+r)^t} \right).$$

Take the derivative with respect to s_t :

$$\frac{\partial \mathcal{L}}{\partial s_t} = \beta^t u'(s_t) - \lambda \frac{1}{(1+r)^t} = 0.$$

So

$$\beta^t u'(s_t) = \lambda \frac{1}{(1+r)^t}.$$

Multiply by $(1 + r)^t$:

$$\beta^t(1 + r)^t u'(s_t) = \lambda.$$

Since λ is constant over time, for any t and $t + 1$,

$$\beta^t(1 + r)^t u'(s_t) = \beta^{t+1}(1 + r)^{t+1} u'(s_{t+1}).$$

$$u'(s_t) = \beta(1 + r) u'(s_{t+1}).$$

Let

$$u(s) = \log s.$$

$$u'(s) = \frac{1}{s}.$$

$$\frac{1}{s_t} = \beta(1 + r) \frac{1}{s_{t+1}}.$$

$$s_{t+1} = \beta(1 + r) s_t.$$

So we see the spending growth rate is

$$\frac{s_{t+1}}{s_t} = \beta(1 + r).$$

This is what is meant by the same consumption rate: the same proportional growth rate over time.

(Rich vs. Poor Player)

In this model, we are going to assume that there will be a rich player (R) and a poor player (P). Since the richer player made great economic decisions in the game, they are projected to have more gold during their lifetime than the poor player.

$$W^R > W^P.$$

Each solves

$$\max_{\{s_t^i\}} \sum_{t=0}^T \beta^t u(s_t^i) \quad \text{subject to} \quad \sum_{t=0}^T \frac{s_t^i}{(1 + r)^t} = W^i, \quad i \in \{R, P\}.$$

The Euler equation for each player is

$$u'(s_t^R) = \beta(1 + r) u'(s_{t+1}^R),$$

and

$$u'(s_t^P) = \beta(1 + r) u'(s_{t+1}^P).$$

Under log utility, this gives

$$\frac{s_{t+1}^R}{s_t^R} = \beta(1+r), \quad \frac{s_{t+1}^P}{s_t^P} = \beta(1+r).$$

So

$$\frac{s_{t+1}^R}{s_t^R} = \frac{s_{t+1}^P}{s_t^P}.$$

Thus, both players have the same spending growth rate.

Step 7: Solving for the Level of Spending

Since

$$s_{t+1} = \beta(1+r)s_t,$$

we can write

$$s_t = s_0[\beta(1+r)]^t.$$

Plug this into the budget constraint:

$$\sum_{t=0}^T \frac{s_0[\beta(1+r)]^t}{(1+r)^t} = W.$$

Simplify:

$$s_0 \sum_{t=0}^T \beta^t = W.$$

So

$$s_0 = \frac{W}{\sum_{\tau=0}^T \beta^\tau}.$$

Therefore the full spending path is

$$s_t = \frac{W}{\sum_{\tau=0}^T \beta^\tau} [\beta(1+r)]^t.$$

For the rich player:

$$s_t^R = \frac{W^R}{\sum_{\tau=0}^T \beta^\tau} [\beta(1+r)]^t,$$

and for the poor player:

$$s_t^P = \frac{W^P}{\sum_{\tau=0}^T \beta^\tau} [\beta(1+r)]^t.$$

Taking the ratio gives

$$\frac{s_t^R}{s_t^P} = \frac{W^R}{W^P}.$$

So, if the rich player has more gold during their lifetime, then they would spend more at all stages, even though both players are going the same spending path (Assuming no deviations from level ups).

(Why Playing the Same Board is Rather Unwise)

A board requiring spending path $\{s_t^*\}$ is feasible for player i only if

$$\sum_{t=0}^T \frac{s_t^*}{(1+r)^t} \leq W^i.$$

So it may be possible for both players to attempt the same board.

$$\boxed{s_t^R > s_t^P \quad \forall t.}$$

This implies the rich player can consistently apply more pressure in contested situations. In TFT terms, this means the rich player can typically

- reroll more often,
- level more aggressively,
- buy more units along the contested line,
- recover from misses more easily.

Thus, even if the poor player can try to play the same board, it may be strategically inferior to do so.

Conclusion

$$\sum_{t=0}^T \frac{s_t}{(1+r)^t} = W,$$

$$\frac{s_{t+1}}{s_t} = \beta(1+r),$$

same growth rate $\neq$ same spending level.

$$W^R > W^P,$$

$$s_t^R > s_t^P \quad \forall t,$$

$$\frac{s_{t+1}^R}{s_t^R} = \frac{s_{t+1}^P}{s_t^P}.$$

So the poor player should not just blindly copy the same board as the rich player just because both follow the same build path in Mobalytics. Due to the fact that the rich player can sustain the same plan at a superior level.